



MMAUG / AI & DEVOPS BOOTCAMP

INTRODUCTION TO DEVOPS

From Waterfall to Agile to DevOps — how modern teams plan, build, ship and run software

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MMAUG AI & DevOps Bootcamp · 2026

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Today's session

A 45-minute journey — from how we used to build software to how the best teams ship it today

Waterfall Model

Agile Model

What is DevOps

DevOps Phases

DevOps Tools

DevOps Advantage

Career in DevOps

Why delivery had to evolve — Waterfall's limits and Agile's answer

What DevOps is: culture, lifecycle, phases and CI/CD explained

The modern DevOps toolchain mapped to the lifecycle

Roles, skills, 2026 UK & US salaries and a learning path

Resources to keep learning after the session



About me

- ADEDEJI AWOLESI
- SENIOR FINOPS ENGINEER (DEVOPS) – CIVICA
- BSC MANAGEMENT INFORMATION SYSTEMS
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What participants will learn

By the end of this 45-minute session, you will be able to...

01

01

Explain

The limits of Waterfall, what Agile changed — and why the Ops gap remained

02

02

Design

A DevOps lifecycle flow from plan to monitor, with CI/CD stages mapped to it

03

03

Build

Your personal DevOps learning roadmap: skills, tools and certifications

04

04

Evaluate

Which DevOps role fits you — and what it pays in the UK and US in 2026





SECTION / 01

The Evolution of Software Delivery

| *Waterfall Model*





The Waterfall Model

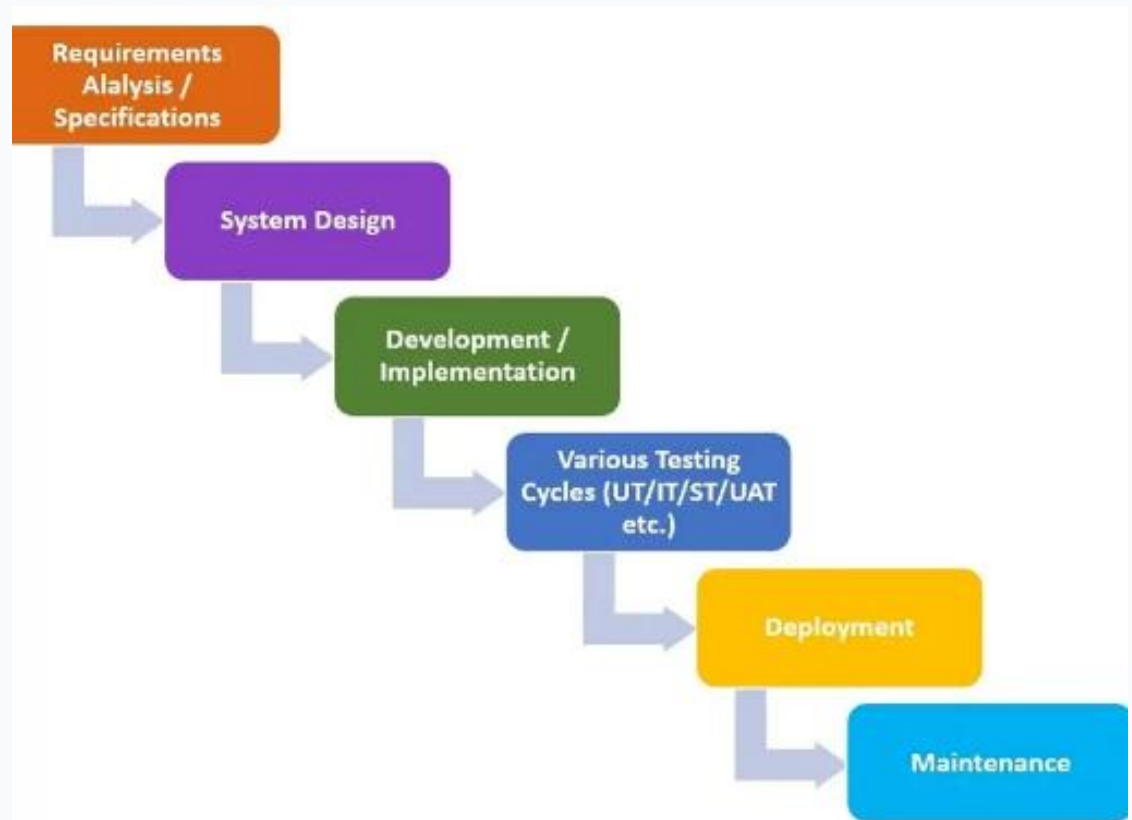
The traditional approach: every phase must finish before the next one begins

A linear, step-by-step approach to software development — phases cascade downwards like a waterfall.

For decades this was how software was built: predictable on paper, painful in practice. Requirements were fixed up front, change became expensive, and users only saw the product at the very end.

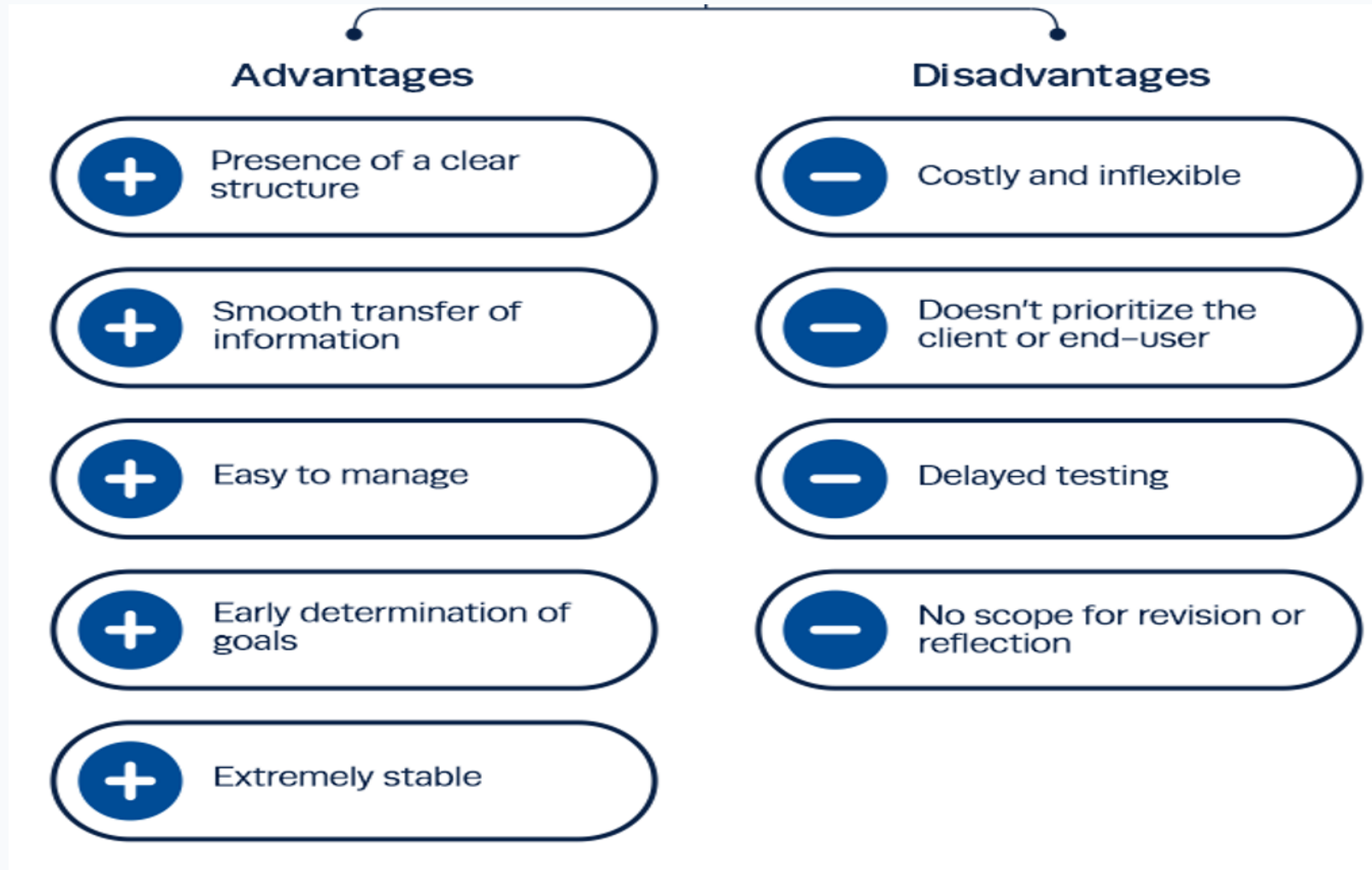
EXAMPLE

A bank spends 18 months building to the original specification — on day one, users realise the reports they actually need were never in the requirements.





The Waterfall Model



The Waterfall Model

Using Water Fall model, Companies now realise that



Client requirements cannot be understood at once



It is very expensive to make changes during the end of the project



Software must be delivered faster and with less resources



The Evolution of Software Delivery (2)

| *Agile Model*

The Agile Model

Small cycles, constant feedback, working software every sprint

Work is broken into small actionable blocks called sprints — each one plans, builds, reviews and adapts in a short cycle.

Programmers create prototypes fast, clients give feedback, and the loop repeats. Direction can change every sprint instead of once a year — the product improves continuously.

Feedback loop

EXAMPLE

A retail site ships a basic checkout in sprint 1; customer feedback adds PayPal in sprint 2 and address autofill in sprint 3.

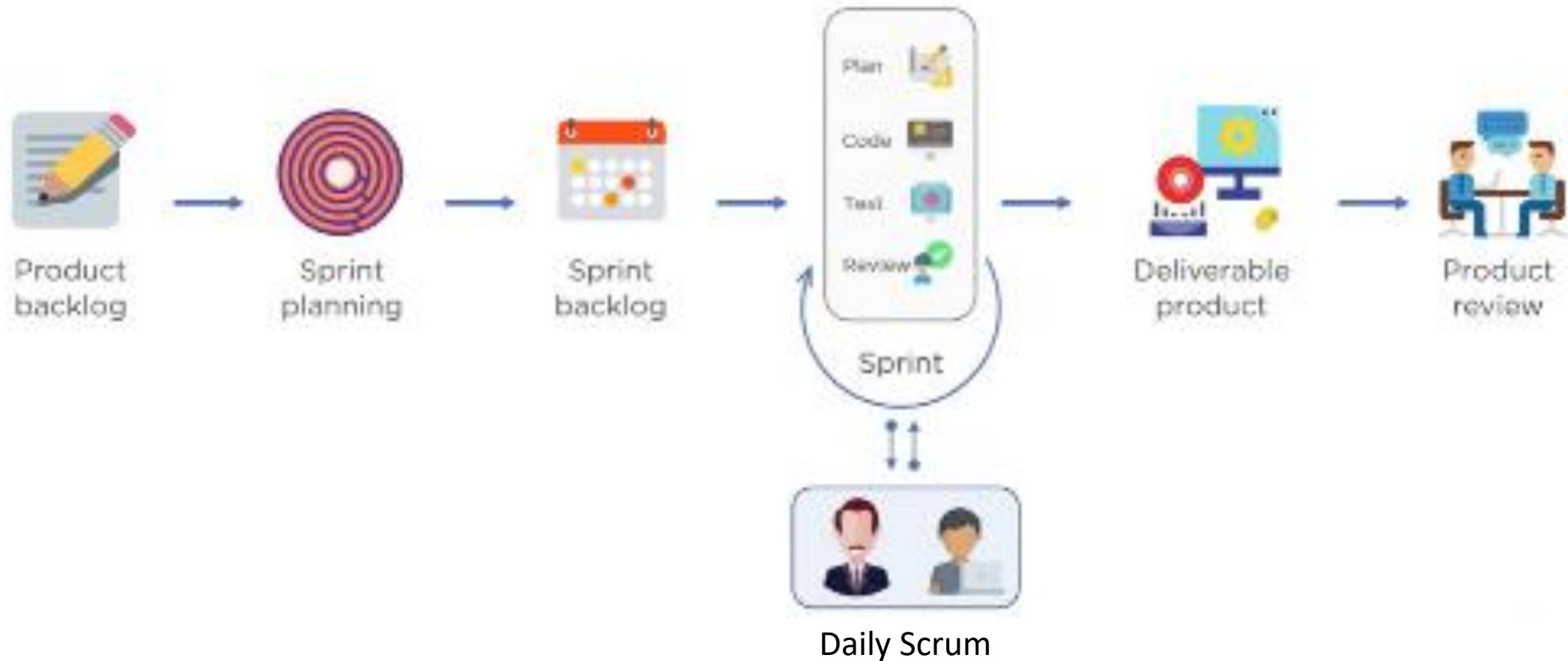


The Agile Model

The entire process of building software is broken down into small actionable blocks called sprints



The Agile Model Workflow





The Agile Model

Advantages	Disadvantages
Clients' requirements are better understood because of constant feedback	Product get tested only on developers' computers and not production systems
Product is delivered faster as compared to waterfall model	Development and Operations work in silos





Waterfall and Agile

Two answers to the same question: how should we organise software delivery?

WATERFALL

- Fit: fixed scope, well-understood requirements
- Advantage: predictable plan, clear documentation
- Trade-off: feedback late; change is costly
- Use case: regulated, fixed-scope projects

AGILE

- Fit: evolving requirements, fast-moving products
- Advantage: sprints and feedback deliver value sooner
- Trade-off: scope creep; needs discipline
- Use case: web apps, startups, product teams

DECISION

Fixed scope with costly change → Waterfall. Learning and feedback matter more than the plan → Agile.





Agile: what it fixed — and what it missed

Agile transformed development, but left a gap between Dev and Ops

WHAT AGILE FIXED

- Requirements understood through constant feedback
- Products delivered faster than Waterfall
- Working software every sprint — visible progress
- Clients involved throughout, not just at the end

WHAT AGILE MISSED

- Tested on developers' machines — not production
- Development and Operations still in silos
- No ownership of deployments and uptime
- The classic excuse: “works on my laptop”

DECISION

Agile sped up development — but the handover to Operations stayed manual. That gap is exactly what DevOps was built to close.





SECTION / 02

Understanding DevOps

One team, one pipeline: culture, lifecycle and CI/CD working together as a single system.



What is DevOps?

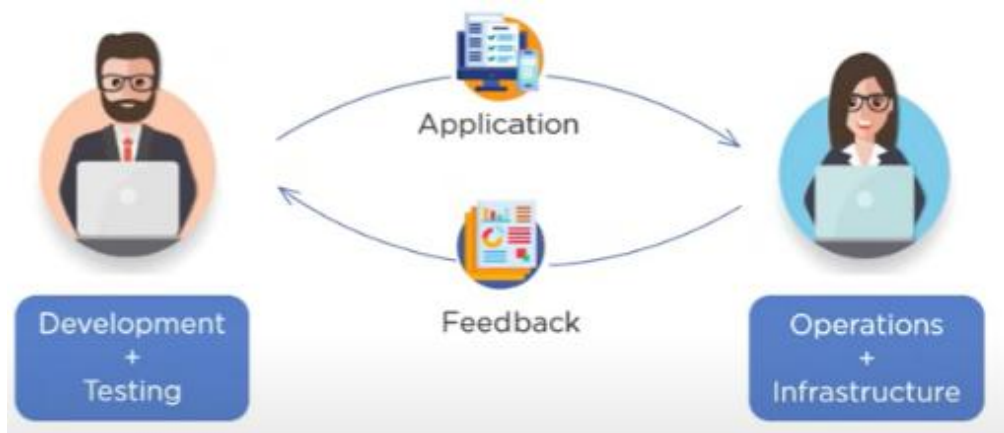
Development + Operations — uniting people, process and technology

Evolution from Agile model of Software Development

Addresses the gap between Developers and Operations



DevOps combines development (Dev) and operations (Ops) to unite people, process and technology in application planning, development, delivery and operations.

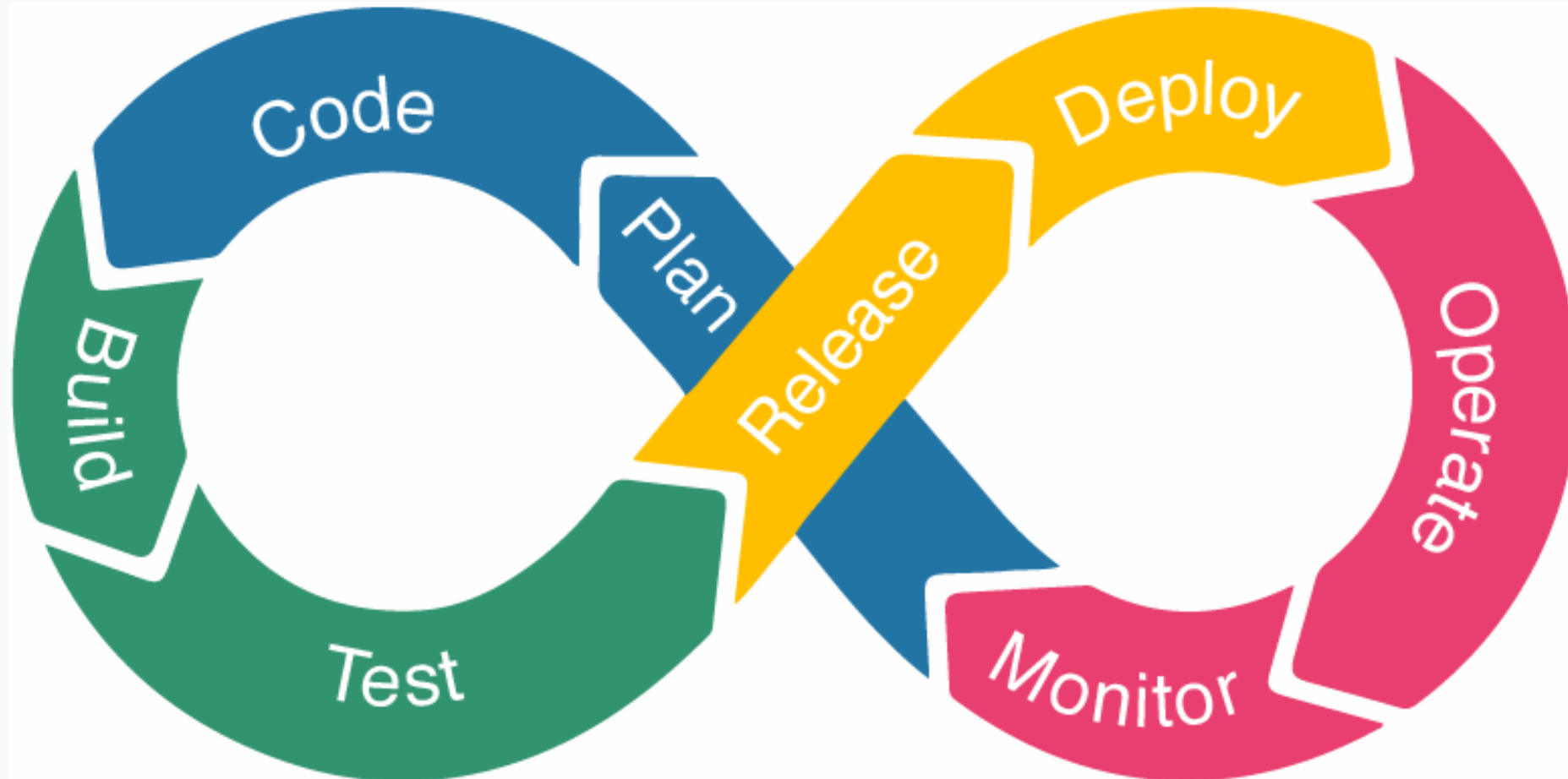


EXAMPLE

A developer's commit triggers an automated pipeline that builds, tests and deploys to staging — and, once approved, rolls out to production the same day.

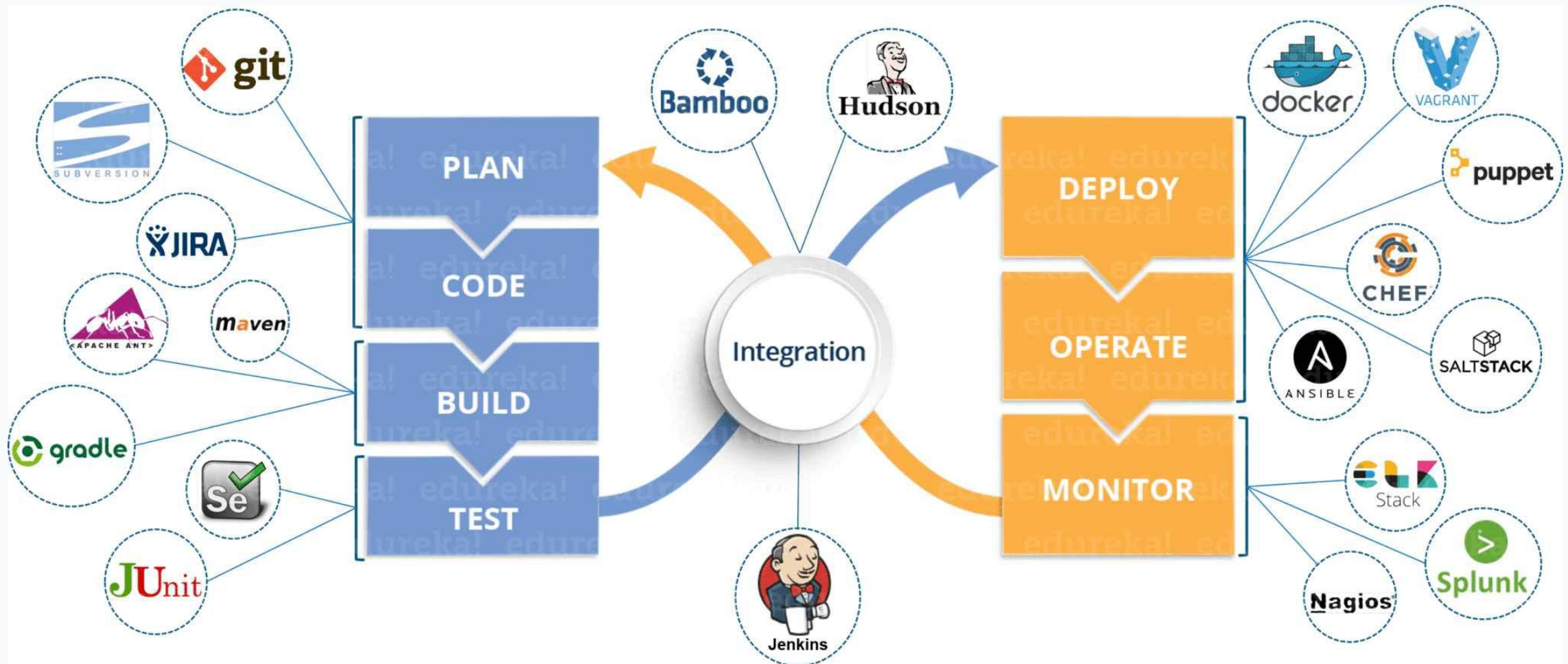
The DevOps Phases

Eight phases, one continuous loop — every phase feeds the next



The DevOps Toolchain

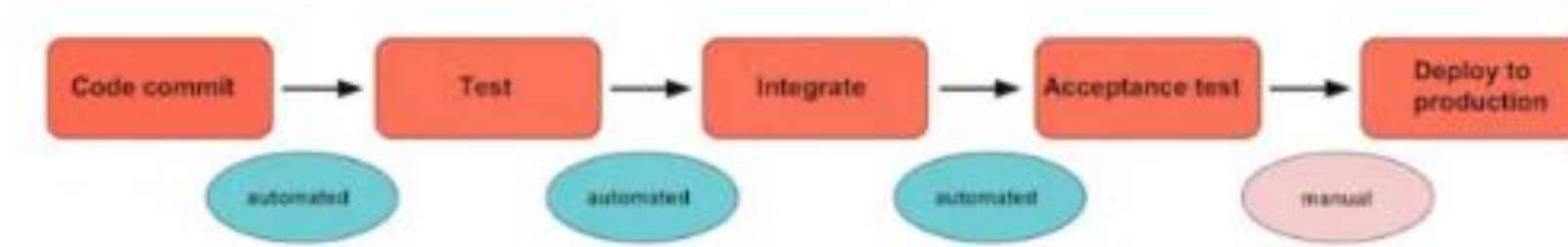
Popular open-source and cloud tools mapped to the lifecycle



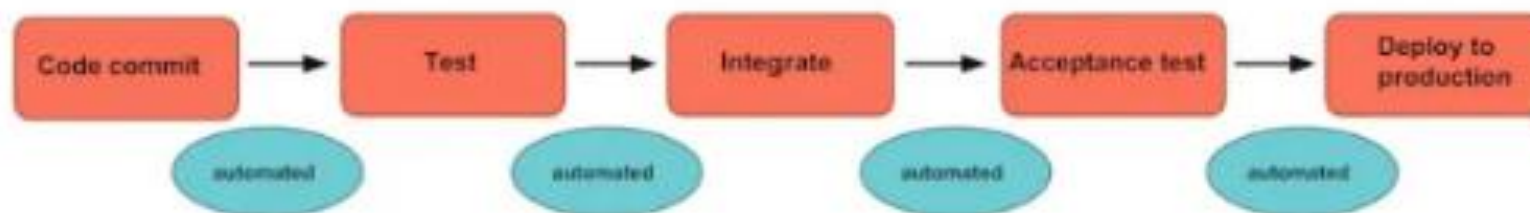
CI/CD: the engine of DevOps

Continuous Integration: Practice in which code changes are automatically built, tested and prepared for release to production in a consistent and reliable manner.

Make sure that project is always in a releasable state but decision to release is approved by development team.



Continuous Deployment: every change that passes through the automated tests and builds is automatically deployed to production without any human intervention.





Continuous Delivery vs Continuous Deployment

Same pipeline — the only difference is one approval gate

CONTINUOUS DELIVERY

- Every change built, tested and kept ready to release
- Advantage: the business decides when to ship
- Trade-off: a manual approval step remains
- Best for: regulated industries, sensitive releases

CONTINUOUS DEPLOYMENT

- Every passing change deploys to production automatically
- Advantage: fastest possible feedback from real users
- Trade-off: demands excellent tests and observability
- Best for: mature teams, SaaS products

DECISION

Keep the approval gate while you build trust in your tests — remove it when your pipeline proves itself.





Why DevOps wins

What organisations gain when Dev and Ops work as one team

01

01

Speed

Faster creation and delivery of products — CI/CD shrinks time-to-market from months to minutes

02

02

Simplicity

Reduced complexity of maintaining applications — small, frequent changes beat big-bang releases

03

03

Collaboration

Improved collaboration between developers and the operations team — shared goals, shared metrics

04

04

Feedback

Faster feedback from users and systems — every deployment teaches the team something

Net effect: deploy in hours not months, recover in minutes, and release value as soon as it is ready.





DevOps in the Real World

The world's best engineering teams run on DevOps principles





SECTION / 03

Building a DevOps Career

Roles, skills, 2026 salaries and a practical learning path to get you started.





Skills You Need for DevOps

Six capability areas — plus Agile ways of working (Scrum, Jira, Azure DevOps)

01

VERSION CONTROL

Git · GitLab — every change tracked and reviewable

02

SCRIPTING

Python · Bash · Go · PowerShell for automation

03

CLOUD

Azure · AWS · GCP — workloads live in the cloud

04

IaC

Terraform · Bicep · ARM · CloudFormation

05

CONTAINERS

Docker · Kubernetes — package and run apps anywhere

06

MONITORING

Prometheus · Grafana · Splunk · Azure Monitor





DevOps Careers: 2026 Salaries

What DevOps roles pay in 2026 — United Kingdom vs United States (indicative, per year)

Role	United Kingdom	United States
Junior DevOps Engineer	£35,000 – £50,000	\$85,000 – \$100,000
DevOps Engineer	£50,000 – £80,000	\$115,000 – \$145,000
Senior DevOps Engineer	£80,000 – £105,000	\$140,000 – \$175,000
DevOps Lead / Architect	£95,000 – £130,000	\$165,000 – \$220,000+
Site Reliability Engineer	£70,000 – £110,000	\$130,000 – \$185,000
Cloud Engineer	£45,000 – £85,000	\$128,000 – \$165,000
DevSecOps / Security Engineer	£60,000 – £110,000	\$145,000 – \$200,000
Release Manager	£50,000 – £85,000	\$100,000 – \$140,000

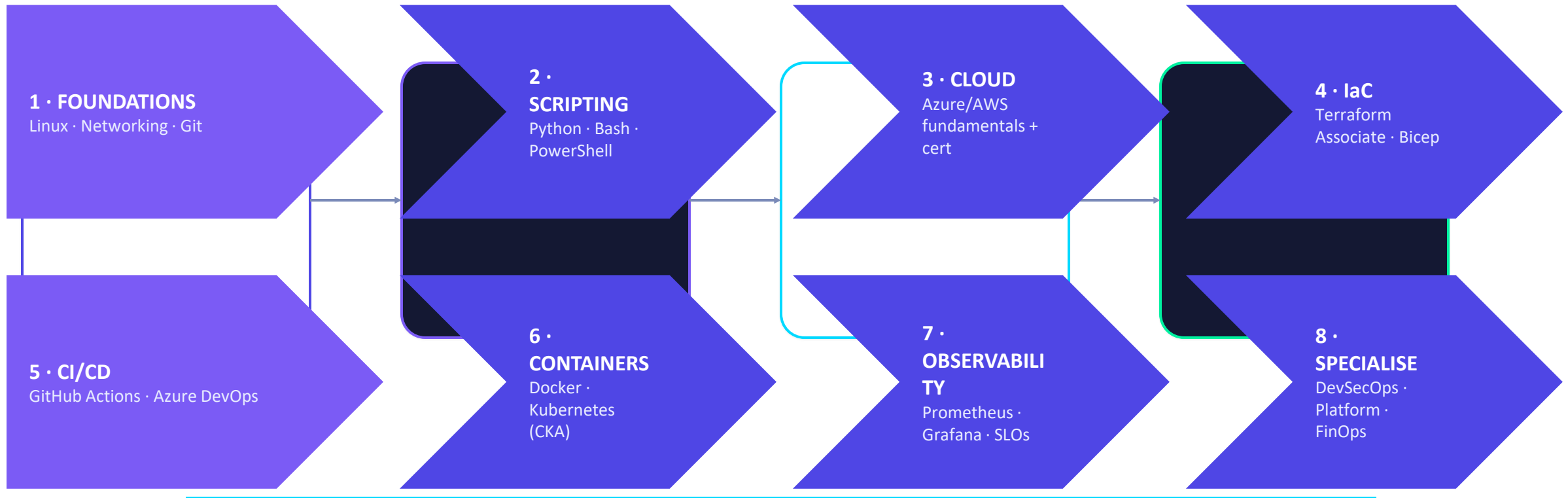
Indicative 2026 market ranges; figures vary with geography, industry and experience — London and US coastal cities sit at the top of each band. Sources: IT Jobs Watch, Glassdoor, ZipRecruiter, Robert Half, KORE1 (2026).





Your DevOps Learning Path

A practical 8-step roadmap from foundations to certification



Rule of thumb: one step at a time — ship a small project at every stage. Part-time, most people reach job-ready in 12–18 months.





What to remember

If you remember only four things from this session...

01

01

Explain

DevOps is culture, not a tool
— people, process and
technology working as one
team

02

02

Design

Automate the path from
commit to production —
CI/CD is the engine, feedback
is the fuel

03

03

Build

Start small: one pipeline, one
service, then scale — and
measure everything

04

04

Evaluate

The career is rich — engineer
to architect, UK to US. Pick a
path and start the roadmap





Continue learning

Give participants direct links to the material they need next

SLIDES	Session deck & blog	adedejiawolesi.com
CODE	Microsoft Learn — DevOps learning paths	learn.microsoft.com/devops
DOCS	GitHub Actions & Azure DevOps docs	learn.microsoft.com/azure/devops
COMMUNITY	MMAUG events and updates	mmaug.com

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